

VARAD ABHIJIT JOSHI

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EDUCATION

University of Southern California

Los Angeles, California

Master of Science in Computer Science

Jan 2024 - Dec 2026

Coursework: Analysis of Algorithms, Natural Language Processing, Multimedia Systems Design, Database Systems

GPA: 3.3/4.0

Mumbai University

Mumbai, India

Bachelor of Engineering in Computer Engineering

Aug 2019 - July 2023

GPA: 3.58 / 4.0

SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript, SQL, Swift
- **Web & Backend :** React, Node.js, Express.js, Django, REST APIs
- **ML & Data :** PyTorch, OpenCV, Pandas, NumPy, MongoDB, MySQL
- **Software Deployment & Tools:** GCP, AWS, Git, Postman

WORK EXPERIENCE

University of Southern California

California, United States

Student Developer @ Facilities Planning & Management / ITS

June 2025 - May 2026

- Cut incident resolution time ~40% by building a real-time integration dashboard using React, TypeScript, Node.js/Express, and MongoDB for AssetWorks–Workday integrations
- Achieved a ~50% reduction in manual updates by automating Python ETL workflows between Workday SOAP XML and AssetWorks REST APIs, while supporting data/API migration and developing automated validation, load, and latency tests

Damacloid Technologies

Mumbai, India

Software Developer Intern

Jan 2022 - Mar 2022

- Developed a React and Formik-based application form with reusable components, advanced validation, and backend API integration
- Simplified validation and future updates by refactoring form logic into reusable React/Formik components and connecting frontend workflows to existing backend APIs

PROJECTS

SkipForward- Multimodal Video Segmentation Engine: Lead AI & Systems Developer | Python, OpenCV, Librosa, OpenAI Whisper, PyTorch, NumPy, DSP

- Developed an end-to-end multimodal pipeline combining 13 visual, acoustic, and NLP feature channels on a unified 1 Hz temporal grid to distinguish core content from ads, intros, and dead air
- Engineered audio feature extraction using RMS energy, spectral flux, spectral flatness, and chroma stability, and integrated local Whisper-based speech-to-text for transcript and semantic analysis

JoshiStocks: Developer | *Swift, SwiftUI, Combine, WebSockets*

- Designed an iOS stock-trading application using Swift and SwiftUI with real-time market data, portfolio management, stock analysis, and transaction workflows
- Implemented WebSocket-based live price updates and interactive UI components for responsive market-data visualization

Healthify: Developer | *Django, React.js, REST APIs, JavaScript* | [Healthify website](#)

- Built full-stack telemedicine workflows for online consultations and healthcare-service access using Django, React.js, and REST APIs
- Connected frontend and backend services through REST APIs and supported testing and bug resolution across core telemedicine workflows

PUBLICATIONS

Real-Enhanced SRGAN for Astronomical Image Super-Resolution:-

- Architected a Real-Enhanced SRGAN for astronomical image super-resolution, achieving ~52% improvement over baseline methods on imagery covering 9,999+ celestial objects
- Peer-reviewed and presented at IEMENTech 2022 (IEEE Kolkata); evaluated against interpolation baselines on deep-space imagery while preserving fine astrophysical structures